

SPEEDING UP REACTIONS

State Of Sub-Division (Size)

- Refers to the size of the pieces of the reactants.
- ***The smaller the pieces, the faster the rate of reaction.***

e.g. Fe in a nail will not burn in a bunsen flame.

Steel wool (which is reasonably open) will burn to some extent in a bunsen flame.

Fe filings are very fine and burn away completely in a bunsen flame.

Smaller pieces increase the effective area available for a reaction to occur.

In the above examples, more O₂ can get to the filings than to the steel wool or nail so they burn very easily.

Concentration Of Reactants

- ***The greater the concentrations of the reactants, the faster the rate of reaction.***

Greater concentrations means that more reactant particles are present.

They can collide more frequently with each other and react to produce product particles.

This is the basis of the Collision Theory.

Collision Theory: If particles collide with sufficient energy to break chemical bonds and form new ones, then a chemical reaction will occur.

e.g. More H₂ gas is produced by adding Zn to 2.0 M HCl (high concentration) rather than 0.5 M HCl (low concentration).

Temperature

- ***The higher the temperature, the faster the rate of reaction.***

At higher temperatures, the reactant particles are moving faster (i.e. have greater kinetic energy) and collide more often.

Hence bonds are broken more easily and the rate increases.

e.g. A Zn/HCl combination will produce H_2 at a faster rate if the beaker is heated gently over a bunsen.

Catalysts

- ***A catalyst is a substance (or mixture of substances) that increases the rate of a reaction without being used up in the reaction.***

e.g. O_2 is produced very quickly if manganese dioxide (MnO_2) is added as a catalyst to hydrogen peroxide (H_2O_2).

After the reaction, the MnO_2 can be seen at the bottom of the reaction vessel - it is unused.